

16.0 Milestone Assessment: Network Systems Characteristics

Introduction — What Makes a Network a System?

Which part of a network system do you think is most crucial to understand for a Network+ certification, and why?

Network systems are the backbone of any IT infrastructure, connecting devices, enabling communication, and supporting data flow across a network. The [CompTIA Network+ Certification Exam Objectives](#) identify five domains that must be understood to successfully complete the Network+ exam. I believe Network Security is especially important today because security must be woven throughout the protocols, devices, services, and processes that make up a network system. At the same time, the success of a Network Engineer depends on understanding and applying concepts from all five domains.

The Network+ domains are:

1. Networking Concepts — 23%
2. Network Implementation — 20%
3. Network Operations — 19%
4. Network Security — 14%
5. Network Troubleshooting — 24%

Ultimately, a Network Engineer must understand how these areas work together to successfully design, build, manage, secure, troubleshoot, and maintain networks of different sizes and complexity. It is essential to understand not only individual components such as routers, switches, firewalls, and wireless access points, but also how those components interact as part of a cohesive network system.

A network system can be understood by following data from the physical connection, through the switched LAN, across routed networks, and ultimately to network services and applications. Security, redundancy, monitoring, troubleshooting, and documentation support the system throughout its lifecycle.

How do you think a well-designed network system impacts business operations and security?

A well-designed network system allows all levels of an organization—from management and employees to authorized guests—to securely access the resources required to perform business functions without having to be concerned with the underlying network infrastructure.

An effective network should support:

- Business continuity
- Data availability, integrity, and security
- Reliable communications
- Performance and scalability
- Secure access to business resources

By understanding how network components, protocols, services, and security controls work together, a Network Engineer can create an enterprise infrastructure with high-performing local and remote connectivity, scalable infrastructure, layered security, and redundant paths and services.

These capabilities support business continuity by allowing employees to maintain access to critical applications, communications, and data as the organization grows and when individual network components or connections fail.

Build the Network from the Bottom Up

1 – Physical Foundation

The physical network must first be evaluated to ensure that the proposed design is feasible, reliable, and scalable. This includes determining the appropriate media – such as copper, fiber, and wireless – and selecting compatible transceivers and network equipment. Fiber connectivity may be required between buildings to support high-speed communication over longer distances, while copper Ethernet cabling must remain within supported distance limitations. High-density areas such as conference rooms and shared workspaces must also be identified because these locations may require additional network capacity and wireless coverage.

Power availability and reliability are also important considerations. Power over Ethernet (PoE) delivers electrical power over Ethernet cabling, allowing compatible switches to power devices such as wireless access points, Voice over IP (VoIP) phones, and cameras. Uninterruptible Power Supplies (UPSs) help protect critical network equipment from temporary power loss, brownouts, and voltage fluctuations while allowing systems to remain operational or shut down gracefully.

Physical security must also protect network infrastructure. Main Distribution Frame (MDF) and Intermediate Distribution Frame (IDF) locations, network equipment, cabling, and supporting power systems should be protected from unauthorized access and tampering.

Finally, the number and types of connected devices – including workstations, VoIP phones, wireless devices, servers, and Internet of Things (IoT) devices – must be estimated. This information helps ensure that the physical network can support both current requirements and future growth.

Physical Network Infrastructure

Physical Component	Key Characteristics	Typical Enterprise Use	Important Considerations
Cat5e	Copper; up to 1 Gbps; 100 m	Existing Ethernet LAN connections	Susceptible to EMI/RFI and crosstalk
Cat6	Copper; 1 Gbps to 100 m; 10 Gbps to ~55 m	Workstations, switches, APs, VoIP phones	Higher performance than Cat5e; distance matters at 10 Gbps
Cat6a	Copper; 10 Gbps to 100 m	Modern enterprise access networks	Good choice for high-speed structured cabling
Single-Mode Fiber (SMF)	Optical; very high bandwidth; several kilometers	Building-to-building, WAN, ISP, long-distance backbone links	Immune to EMI; supports long distances
Multimode Fiber (MMF)	Optical; high bandwidth; hundreds of meters	LAN backbones, data centers, switch uplinks	Immune to EMI; intended for shorter distances than SMF
RJ-45	Copper Ethernet connector	PCs, switches, routers, APs, patch panels	Common with Cat5e/Cat6/Cat6a
LC Fiber Connector	Small-form-factor fiber connector	Enterprise fiber and high-density installations	Commonly paired with SMF/MMF and SFP transceivers
SFP/SFP+ Transceiver	Modular network interface/transceiver	Fiber or compatible copper switch/router uplinks	Transceiver, media, wavelength, distance, and device must be compatible
PoE	Carries electrical power over Ethernet cabling	APs, VoIP phones, cameras, IoT devices	Verify switch PoE capability and available power budget
UPS	Backup and conditioned electrical power	Switches, routers, firewalls, servers, network rooms	Helps maintain availability during power disturbances
MDF / IDF	Central and intermediate network distribution locations	Houses switching, patch panels, cabling, and supporting equipment	Physical security, cooling, power, labeling, and cable management
Wireless / RF	Uses radio frequencies instead of physical cabling to the client	Laptops, mobile devices, IoT, WLAN access	Coverage, interference, channel planning, capacity, and AP placement

2 — Switched Network

Once the physical infrastructure is established, Layer 2 switching provides connectivity between devices within the Local Area Network (LAN). Technologies such as VLANs, access ports, IEEE 802.1Q trunks, Rapid Spanning Tree Protocol (RSTP), Link Aggregation Control Protocol (LACP), port security, and IEEE 802.1X provide segmentation, connectivity, redundancy, link aggregation, and access control.

VLAN Segmentation and Access Ports

Virtual Local Area Networks (VLANs) logically divide a switched LAN into separate Layer 2 broadcast domains. Endpoint devices can therefore be grouped according to department, function, or security requirements while continuing to share the same physical switching infrastructure.

Access ports normally connect endpoint devices to a single VLAN. For example, a workstation connected to an access port assigned to the Finance VLAN becomes part of that VLAN's Layer 2 broadcast domain. Separate VLANs may be created for:

- Human Resources
- Finance
- Sales
- Information Technology
- Servers
- Voice
- Network management
- Guest access

A dedicated Guest VLAN can isolate guest devices from internal systems and sensitive business resources. VLAN segmentation reduces broadcast domains, organizes network traffic, improves security boundaries, and simplifies troubleshooting and policy enforcement.

IEEE 802.1Q Trunking

VLANs often need to extend across multiple switches. IEEE 802.1Q trunk links allow traffic from multiple VLANs to traverse a single physical connection between network devices.

For example, fiber uplinks between switches or buildings can provide high bandwidth and support longer distances while 802.1Q tagging identifies the VLAN associated with Ethernet traffic crossing the trunk. Together, access ports and trunk ports perform different but complementary functions: access ports connect endpoints to their assigned VLAN, while trunk ports transport multiple VLANs between network devices.

Redundancy, RSTP, and LACP

A reliable network should avoid single points of failure. Redundant uplinks between critical switches provide alternate paths if a primary connection becomes unavailable. However, redundant Layer 2 paths can create switching loops, resulting in broadcast storms, unstable Media Access Control (MAC) address tables, duplicate frames, and serious network performance problems.

Rapid Spanning Tree Protocol (RSTP) prevents these loops by maintaining a loop-free Layer 2 topology while preserving redundant paths for failover. The network can also be designed with a predictable root bridge so traffic follows an intentional switching topology. Link Aggregation Control Protocol (LACP) can further improve resiliency and capacity by combining multiple compatible physical links into a single logical link. This provides additional bandwidth while allowing traffic to continue across remaining member links if an individual link fails.

Layer 2 Access Control

Security should also begin at the network access layer. Port security can restrict which MAC addresses are permitted to use a switch port, helping prevent unauthorized devices from connecting through unused or protected access ports while IEEE 802.1X provides identity-based network access control. Working with an authentication service such as RADIUS, 802.1X can require a user or device to authenticate before receiving authorized network access.

3 — Routed Network

VLANs provide Layer 2 segmentation, but devices in different VLANs require Layer 3 routing to communicate. A Layer 3 switch can provide high-speed inter-VLAN routing within the switching infrastructure. A Switch Virtual Interface (SVI) can provide a Layer 3 interface and default gateway for devices within a VLAN. Access Control Lists (ACLs) or firewall policies can then restrict which networks and services are permitted to communicate across these Layer 3 boundaries.

Within larger routed environments, routers and Layer 3 switches use routing tables to determine how packets should reach remote networks. Routes can be manually configured as static routes or dynamically learned through routing protocols.

Wide Area Network

A Wide Area Network (WAN) connects geographically separated networks such as headquarters, branch offices, and data centers. Routers operate at Layer 3 and forward packets between networks according to routing information.

At larger scales, Border Gateway Protocol (BGP) provides path-vector routing and exchanges routing information associated with Autonomous Systems (ASes). BGP uses path attributes and

routing policies when selecting routes, making it well suited to large-scale interdomain routing such as that used throughout the Internet.

- **eBGP** exchanges BGP routing information between different autonomous systems.
- **iBGP** distributes BGP routing information between BGP routers within the same autonomous system.

Protocol	Type	Primary Metric / Decision	Network+ Concept
RIP	Distance-vector IGP	Hop count	Simple routing protocol with a maximum of 15 usable hops
OSPF	Link-state IGP	Cost	Builds a topology database and uses Shortest Path First (SPF)
EIGRP	Advanced distance-vector IGP	Composite metric, primarily bandwidth and delay	Uses DUAL and supports rapid convergence
BGP	Path-vector / interdomain	Path attributes and routing policy	Exchanges routing information associated with autonomous systems

4 — Wireless, Remote, and Cloud Connectivity

Wireless Local Area Network

A Wireless Local Area Network (WLAN) extends enterprise LAN connectivity to laptops, mobile devices, scanners, and other wireless endpoints while allowing users to remain mobile throughout the facility. IEEE 802.11 Wi-Fi provides wireless connectivity, while proper Wireless Access Point (AP) placement, channel planning, coverage, and capacity planning help maintain performance in areas with different user densities.

An access point bridges wireless clients into the wired network. Service Set Identifiers (SSIDs) can be associated with appropriate VLANs so that different classes of wireless traffic remain logically separated. For example, corporate wireless users can be associated with an internal VLAN while guest wireless users are placed into a dedicated Guest VLAN.

Corporate and Guest Wireless Networks

Wireless Protected Access WPA2/WPA3-Enterprise can use IEEE 802.1X with a Remote Authentication Dial-In User Service (RADIUS) server to provide identity-based network access control. Users or devices authenticate through RADIUS before receiving authorized network access and RADIUS provides centralized Authentication, Authorization, and Accounting (AAA). Additional access points can extend coverage and increase capacity as the organization grows, while overlapping coverage and redundant supporting infrastructure can improve availability.

A separate guest SSID can be mapped to a dedicated Guest VLAN. Guest traffic can then be restricted to Internet access while preventing access to internal business systems. Additional policies such as bandwidth limitations or a captive portal may also be applied.

Wireless Standards and Performance

Modern wireless technologies such as Wi-Fi 6 and Wi-Fi 6E can improve performance and capacity in high-density environments. Appropriate frequency bands, channel planning, signal coverage, interference management, and client compatibility must be considered when designing the WLAN.

Virtual Private Network

Virtual Private Networks (VPNs) provide encrypted connectivity across public or otherwise untrusted networks. VPNs can support remote employees as well as site-to-site connectivity between enterprise locations.

Encryption protects data in transit, while authentication and Multi-Factor Authentication (MFA) help ensure that only authorized users gain remote access. Redundant VPN gateways and Internet connections can further strengthen availability.

Cloud Networking

Cloud networking extends enterprise infrastructure into virtual environments where applications, networks, security controls, storage, and other services can be provided without depending entirely on on-premises hardware. Cloud networking demonstrates the relationship between performance, scalability, security, and reliability:

- Distributed resources and load balancing can reduce individual resource bottlenecks.
- Virtual networks and computing resources can expand or contract as business requirements change.
- Identity and Access Management (IAM), security groups, encryption, segmentation, and virtual firewalls provide layered security.
- Availability zones, replication, backups, load balancing, and disaster recovery strategies can improve service availability.

5 — Centralized Network Services

A connected and routed network still requires network services to become useful to users and applications. Enterprise administrators must understand the services traveling across the infrastructure so switches, routers, firewalls, monitoring systems, and troubleshooting tools can be configured correctly.

Common Enterprise Ports, Protocols, and Services

Protocol / Service	Port(s)	Transport	Enterprise Function
FTP	20/21	TCP	File Transfer Protocol Data/Control
SSH/SFTP	22	TCP	Secure remote administration
Telnet	23	TCP	Remote administration insecure
SMTP	25		Simple Mail Transfer Protocol, Email send
DNS	53	TCP/UDP	Hostname resolution
DHCP	67/68	UDP	Automatic IPv4 configuration Server/Client
HTTP	80	TCP	Web communication
POP3	110	TCP	Email download
IMAP	143	TCP	Internet Message Access Protocol (IMAP)
NTP	123	UDP	Time synchronization
SNMP	161/162	UDP	Network monitoring and alerts
LDAP	389	TCP/UDP	Lightweight Directory Access Protocol (LDAP)
HTTPS	443	TCP	Secure web and application communication
SMB	445	TCP	File and printer sharing
SMTP	587	TCP	Submission, Mail server communication
RDP	3389	TCP/UDP	Remote Windows desktop access

- **TCP** is connection-oriented and provides sequencing, acknowledgments, retransmission, and reliable delivery. It is appropriate for applications where reliable and complete delivery is important.
- **UDP** is connectionless and introduces less overhead. It is useful for services where efficiency or low latency may be more important than retransmitting every lost packet.

IPv6 clients may also obtain addressing information through DHCPv6 or Stateless Address Autoconfiguration (SLAAC). Centralized directory services such as Active Directory can provide authentication, authorization, and centralized management for enterprise users and systems while relying on multiple underlying network protocols and services.

Network Devices Working Together

Enterprise network devices perform specialized but complementary roles.

Device / Appliance	Primary Function	Typical OSI Operation	Enterprise Role / Example
Layer 2 Switch	Connects devices within LANs and VLANs	Layer 2	Connects PCs, phones, printers, and APs
Wireless Access Point	Provides wireless network connectivity	Layers 1–2	Connects wireless clients to the wired LAN
Layer 3 Switch	Combines switching and routing	Layers 2–3	Provides high-speed inter-VLAN routing
Router	Connects separate IP networks	Layer 3	Routes traffic toward WANs and external networks
VPN Gateway	Establishes encrypted VPN connectivity	Multiple layers	Supports secure remote-access and site-to-site connectivity
Firewall / NGFW	Filters and inspects traffic according to policy	Multiple layers	Protects network boundaries and security zones
IDS / IPS	Detects or prevents malicious network activity	Multiple layers	Provides threat detection and prevention
Load Balancer	Distributes application requests	Layers 4 / 7	Improves application performance and availability
DHCP Server	Provides IP configuration	Application layer service	Configures network clients
DNS Server	Resolves names to IP addresses	Application layer service	Allows clients to locate network resources

These devices work as parts of one system rather than as isolated technologies. For example, a laptop may connect through a wireless AP, enter an assigned VLAN, receive IP configuration from DHCP, use DNS to locate a service, cross a Layer 3 switch or router to reach another network, and pass through a firewall before accessing an Internet or cloud resource. Each component performs a different function, but together they provide end-to-end network communication.

6 — Security Across the Network

A secure enterprise network relies on defense-in-depth, where multiple administrative, physical, and technical controls work together to protect organizational resources. Rather than depending on a single security device, security should be incorporated throughout the network architecture.

Security Area	Recommended Controls	Enterprise Benefit
Physical Security	Secure MDF/IDF rooms, environmental controls, access logging	Protects network infrastructure from unauthorized physical access
Identity & Access Management	Directory services, MFA, IEEE 802.1X, Principle of Least Privilege	Verifies identities and limits unauthorized access
Network Security	NGFW, IDS/IPS, VLANs, ACLs, encrypted VPNs	Protects traffic and separates security zones
System & Administrative Security	Patch management, backups, disaster recovery, security policies and awareness	Reduces vulnerabilities and supports recovery and continuity
Monitoring & Response	SNMP, Syslog, packet analysis, SIEM, EDR	Supports detection, investigation, and response

Together, these layered controls complement VLAN segmentation, centralized network services, wireless security, routed boundaries, and redundant infrastructure to provide a secure and manageable enterprise network.

Security therefore becomes a property of the entire network system rather than a function performed only by the firewall.

7 — Operations & Documentation

Building the network is only the beginning of its lifecycle. A production network must be continuously monitored, validated, maintained, troubleshot, and documented so that it remains reliable and understandable as the environment changes.

Monitor

Centralized logging and monitoring provide visibility into network performance, availability, and security events. SNMP can monitor device status and performance information, while Syslog can centralize messages generated by routers, switches, firewalls, servers, and other infrastructure.

Performance baselines can help administrators recognize abnormal changes in utilization, latency, errors, and other network conditions.

Validate

After implementation or configuration changes, the network should be tested to verify that it operates as intended. Validation can include:

- Testing endpoint connectivity

- Verifying VLAN assignments and isolation
- Confirming inter-VLAN routing
- Testing DHCP and DNS functionality
- Validating wireless connectivity and guest restrictions
- Testing redundant links and failover
- Confirming ACL and firewall policies
- Measuring latency, throughput, and availability

Troubleshoot

Network troubleshooting should follow a systematic process rather than making unplanned configuration changes.

A structured approach includes identifying the problem, gathering information, establishing and testing a theory of probable cause, creating and implementing a plan of action, verifying full system functionality, and documenting the findings and solution.

Troubleshooting tools such as ping, traceroute/tracert, ipconfig/ifconfig, nslookup, packet capture tools, interface statistics, logs, and network monitoring platforms can help isolate problems at different layers of the network.

Maintain and Document

Accurate maintenance and documentation keep a network reliable, secure, recoverable, and manageable as the environment changes. Ongoing maintenance includes applying approved updates, backing up configurations, reviewing logs, monitoring performance and capacity, testing redundancy and recovery procedures, and following change-management processes.

Documentation should be updated whenever the network changes and include:

- Physical and logical diagrams,
- IP addressing and subnet plans,
- VLAN and interface assignments,
- Routing information,
- Device inventories,
- Configuration backups,
- Security policies,
- Change records, and
- Troubleshooting procedures.

Together, maintenance and documentation provide Network Engineers with a reliable record of how the infrastructure is designed, how it should operate, and how to efficiently troubleshoot, modify, or recover the network.

Business Impact / Conclusion

A well-designed network system is more than a collection of switches, routers, protocols, and security appliances. It is an integrated infrastructure in which each component contributes to the organization's ability to communicate, access information, protect resources, and continue operating.

The physical infrastructure provides reliable connectivity and power. Layer 2 switching and VLANs organize and segment local traffic. Layer 3 routing connects separate networks. Wireless, VPN, WAN, and cloud technologies extend connectivity beyond the traditional wired LAN. Centralized services such as DHCP and DNS make network resources usable, while layered security controls protect users, devices, traffic, and data. Monitoring, troubleshooting, maintenance, and documentation ensure that the network continues to operate as designed.

Together, these technologies support four major enterprise objectives:

- **Performance:** Efficient switching, routing, appropriate media, network services, and traffic management support responsive applications and communications.
- **Scalability:** Hierarchical design, VLAN segmentation, wireless expansion, WAN connectivity, and cloud resources allow the infrastructure to grow with the organization.
- **Security:** Physical controls, segmentation, authentication, ACLs, firewalls, encryption, monitoring, and defense-in-depth protect business resources.
- **Reliability:** Redundant links, RSTP, LACP, UPS protection, monitoring, backups, and recovery processes reduce single points of failure and support continued operation.

Ultimately, successful network engineering requires understanding both the individual technologies and the relationships between them. Building the network establishes connectivity; securing, monitoring, troubleshooting, maintaining, and documenting it turns that connectivity into a dependable network system capable of supporting long-term business operations.

<https://herbal-rancher.github.io/network-portfolio/>

References

1. **CompTIA.** *CompTIA Network+ Certification Exam Objectives (N10-009), Version 9.*
<https://lecbyo.files.cmp.optimizely.com/download/35a7403ab73211ef9dcda6f347fbf652>
2. **Sanders-Scott, Sherri.** *4.0 Milestone Assessment: Network Troubleshooting — Network Incident Investigation.*
<https://herbal-rancher.github.io/network-portfolio/formative-modules/lesson-4-0/milestone-network-incident-investigation/>
3. **Sanders-Scott, Sherri.** *5.0 Milestone Assessment: Network Infrastructure Design Proposal.*
<https://herbal-rancher.github.io/network-portfolio/formative-modules/lesson-5-0/milestone-network-infrastructure-design/>
4. **Sanders-Scott, Sherri.** *6.0 Milestone Assessment: Enterprise Network Technologies.*
<https://herbal-rancher.github.io/network-portfolio/formative-modules/lesson-6-0/milestone-enterprise-network-technologies/>
5. **Sanders-Scott, Sherri.** *11.0 Milestone Assessment: Network Topologies.*
<https://herbal-rancher.github.io/network-portfolio/formative-modules/lesson-11-0/milestone-network-topologies/>
6. **Sanders-Scott, Sherri.** *12.0 Milestone Assessment: Cabling and Physical Network Infrastructure.*
<https://herbal-rancher.github.io/network-portfolio/formative-modules/lesson-12-0/milestone-network-cabling/>
7. **Sanders-Scott, Sherri.** *15.0 Milestone Assessment: Network Routing Protocols, Segmentation, and Tracing.*
<https://herbal-rancher.github.io/network-portfolio/formative-modules/lesson-15-0/milestone-routing-network-segmentation-tracing/>